

CS 180 Discussion 10

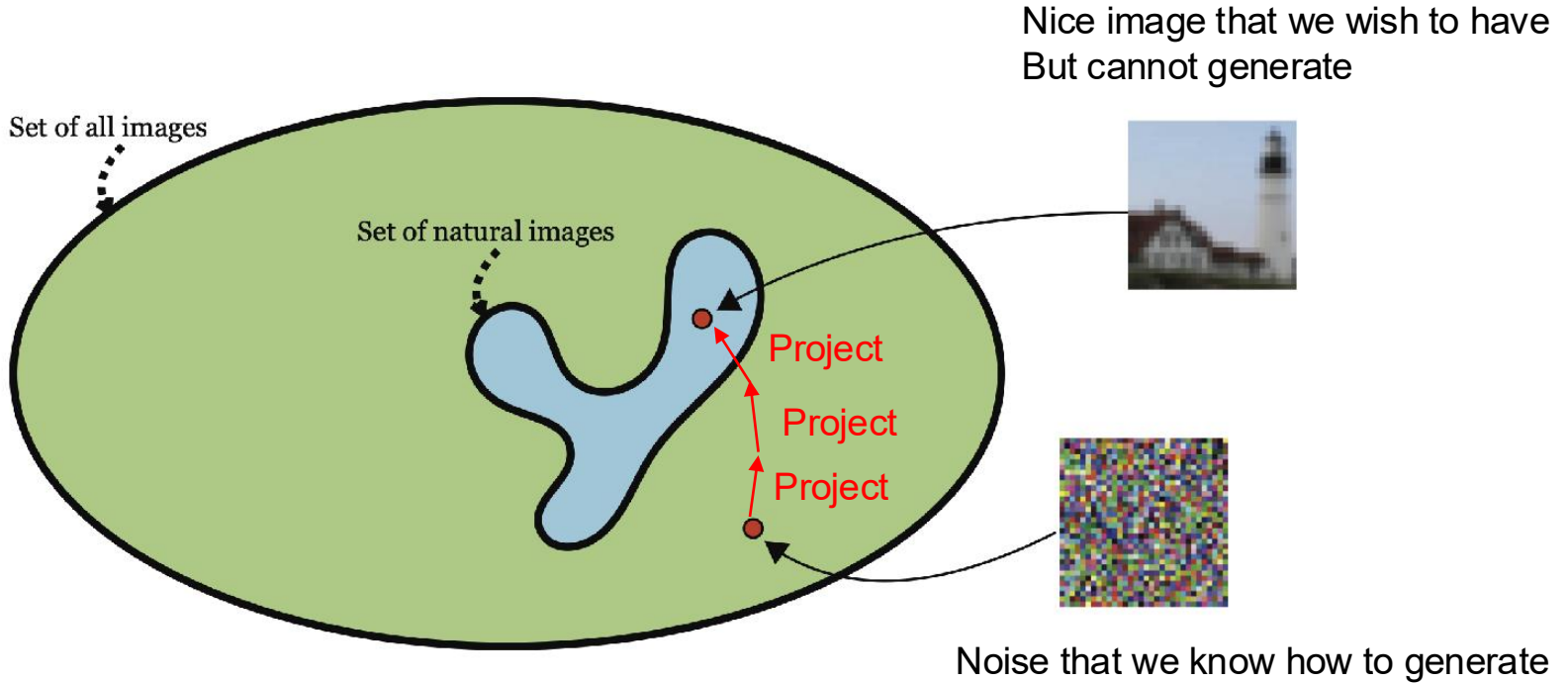
Diffusion and Flow matching

Agenda

1. How diffusion / flow matching works?
2. How CFG works?

**Generating images by
“projecting” onto the manifold**

The natural image manifold



**Diffusion process makes
the “project” possible!**

Diffusion process makes the “project” possible!

Forward (adding noise)



Diffusion process makes the “project” possible!

Reverse (denoising / generating)



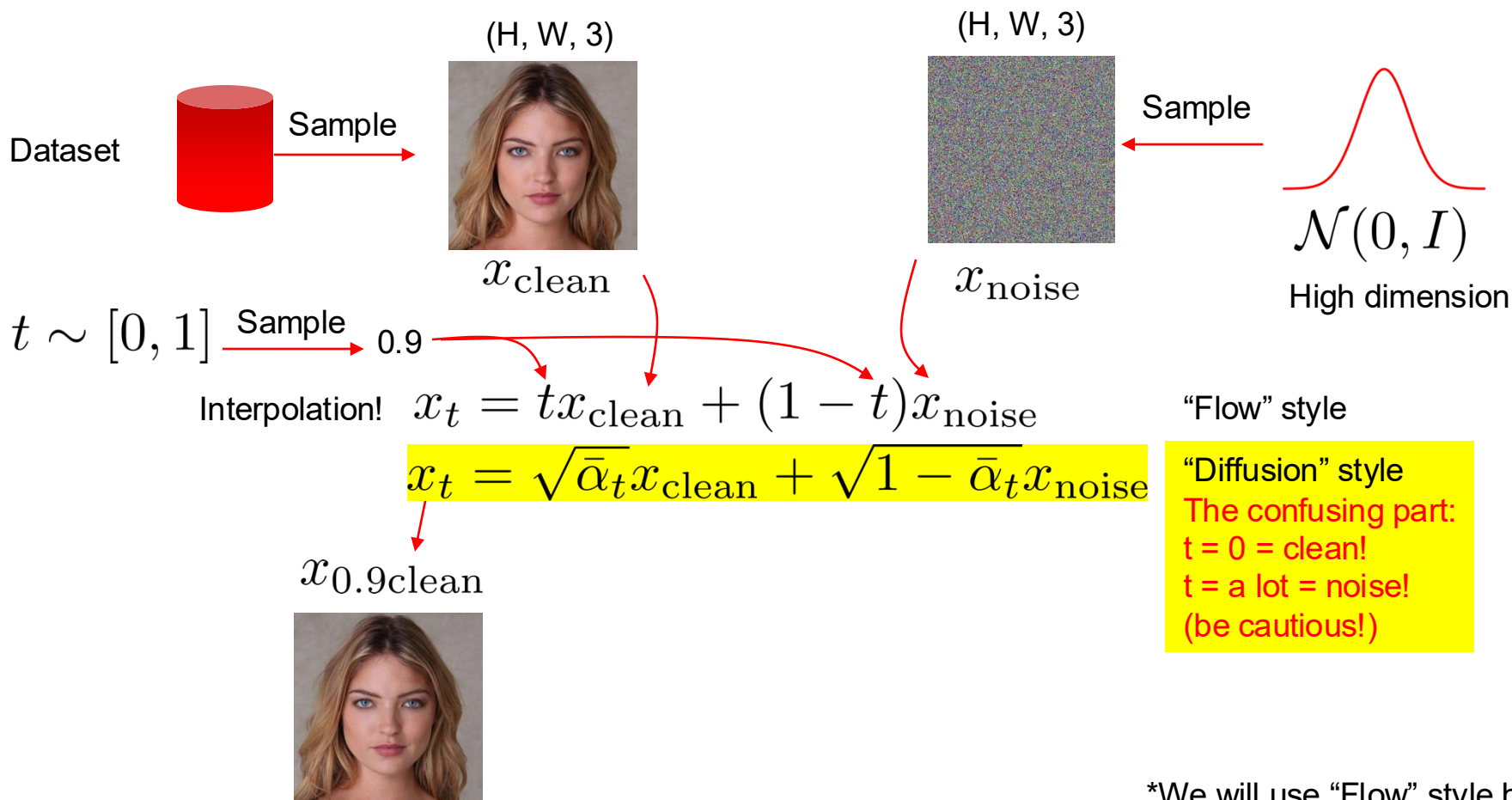
How to properly add noise and denoise?

(If it's that easy we wouldn't wait until 2020 to make it work)

How to add noise?

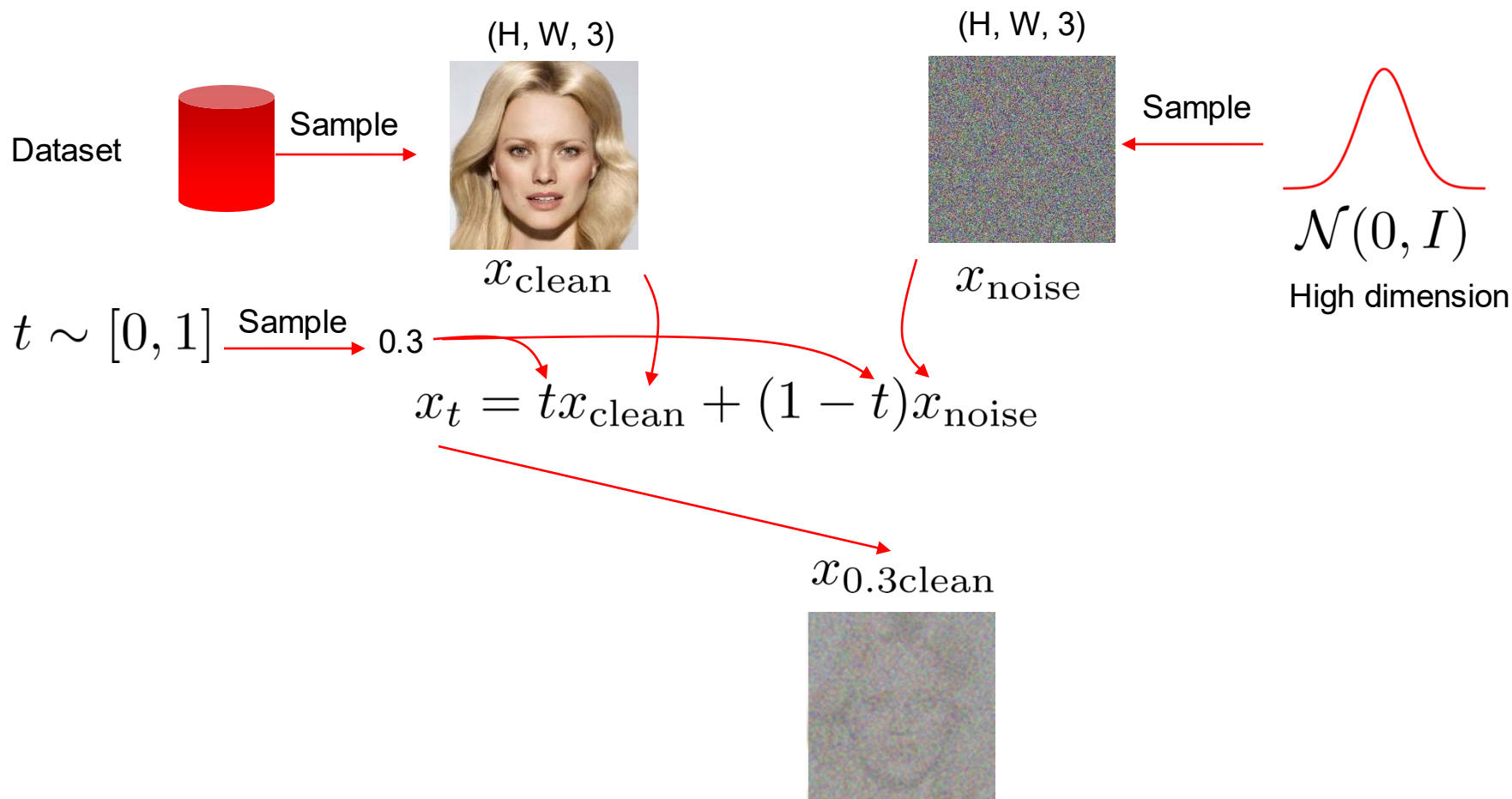


How to add noise?



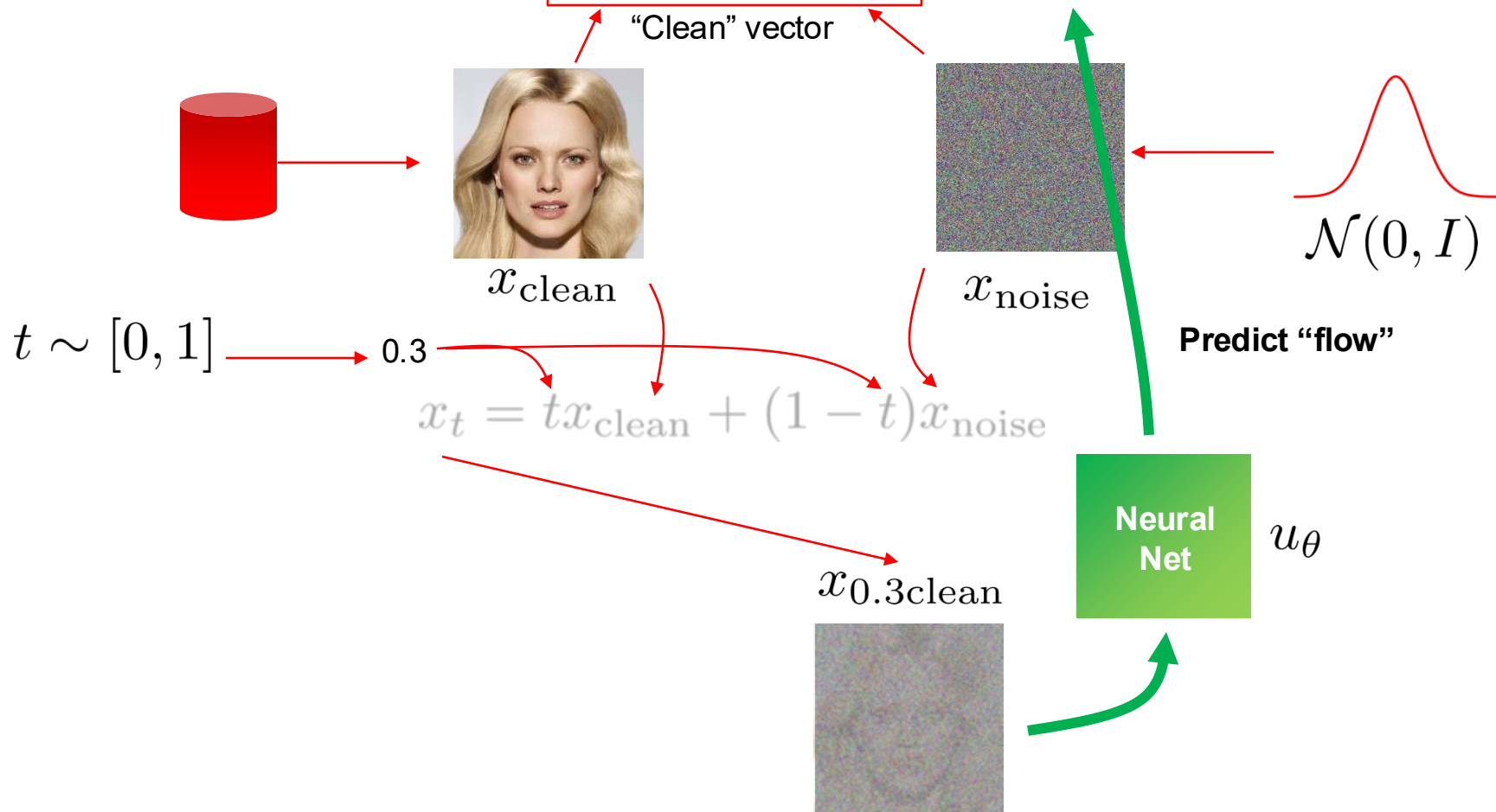
*We will use "Flow" style here!

How to add noise?

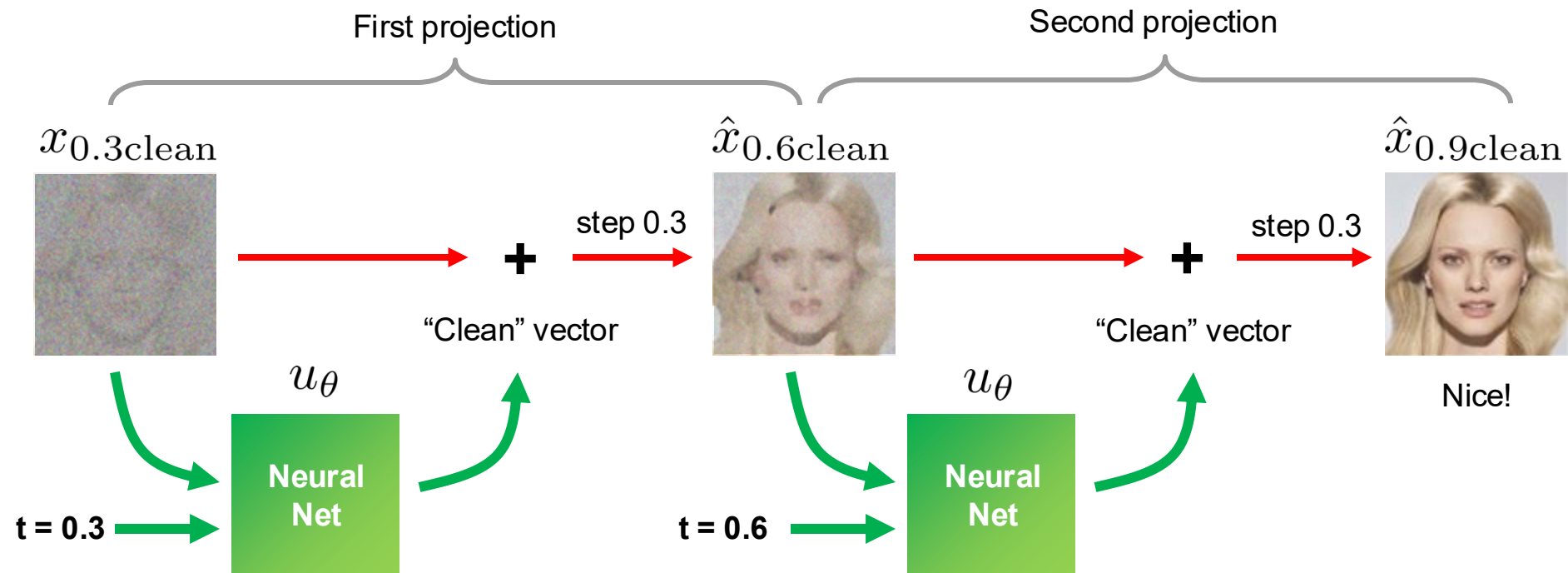


Learn to denoise...

Flow matching loss: $L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_t)||^2$



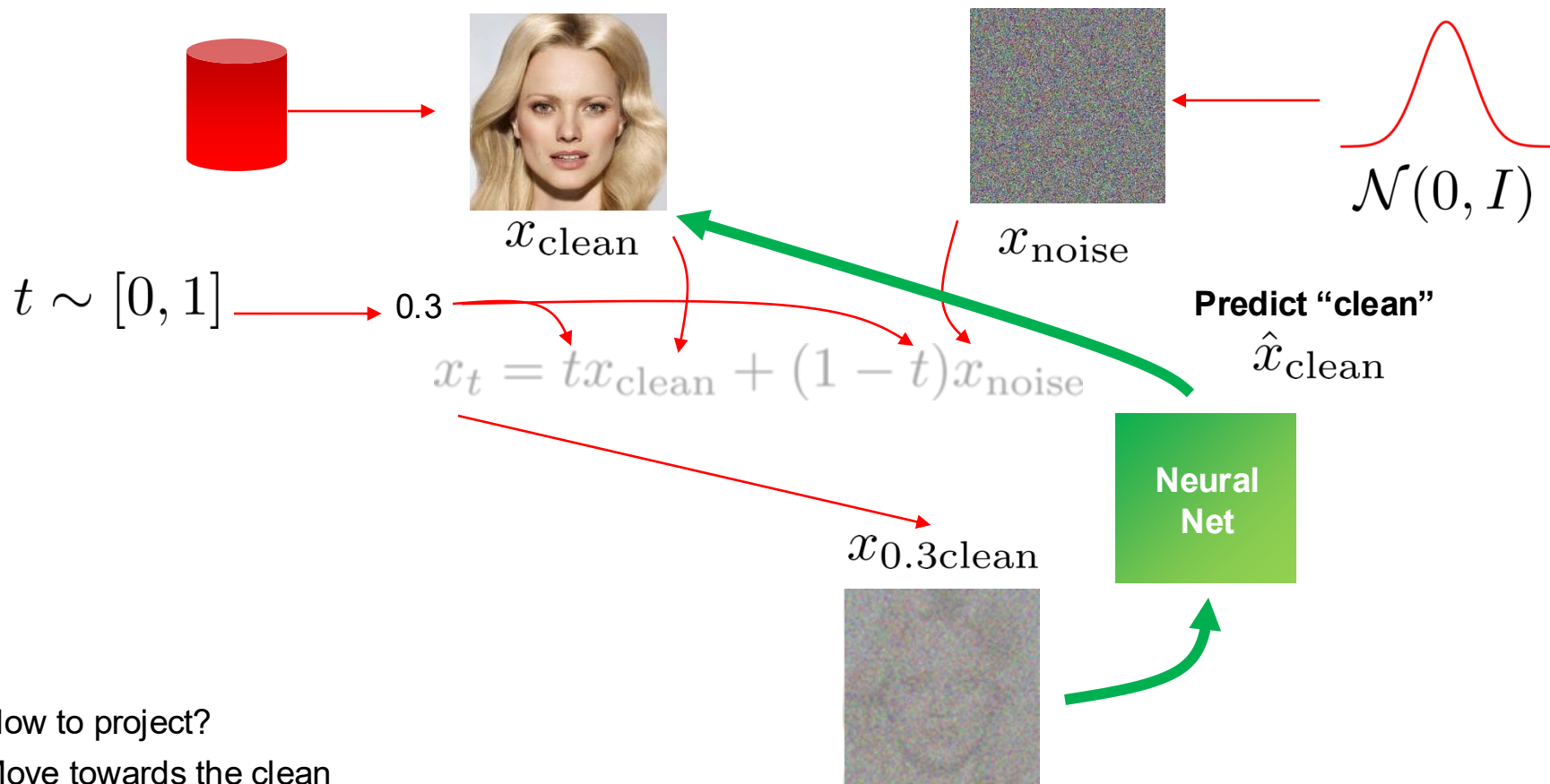
Now, how to project? Ideas?



*Subtle but useful

*Adjustable step size

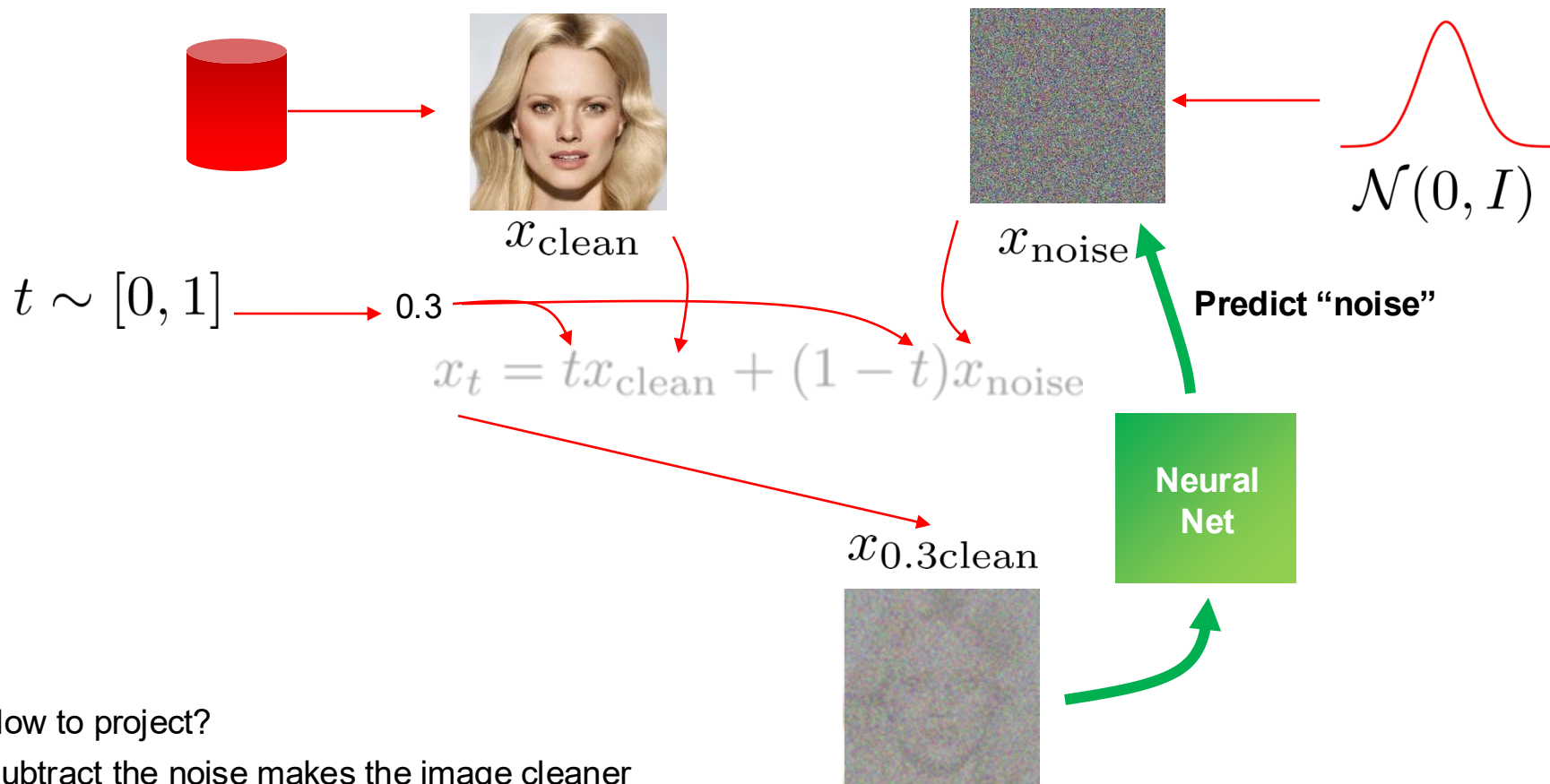
But, you can also predict ... something else ...



*How to project?

Move towards the clean

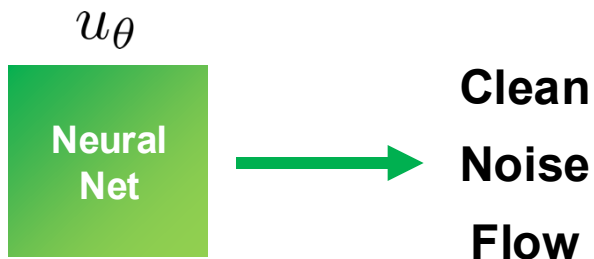
But, you can also predict ... something else ...



*How to project?

Subtract the noise makes the image cleaner

Whatever you choose to predict:



There is an equation* for any sampling
(Just follow the right equation!)

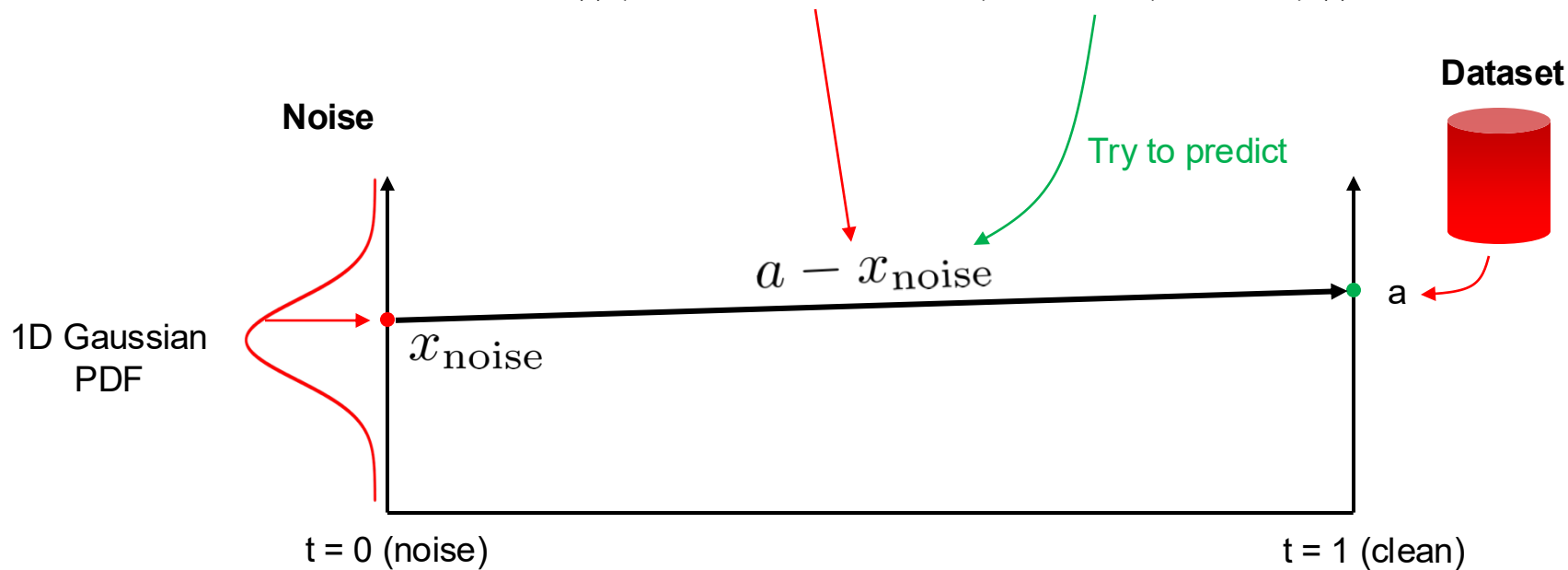
**To really understand,
Imagine a 1D world**

1D world with 2 datapoints



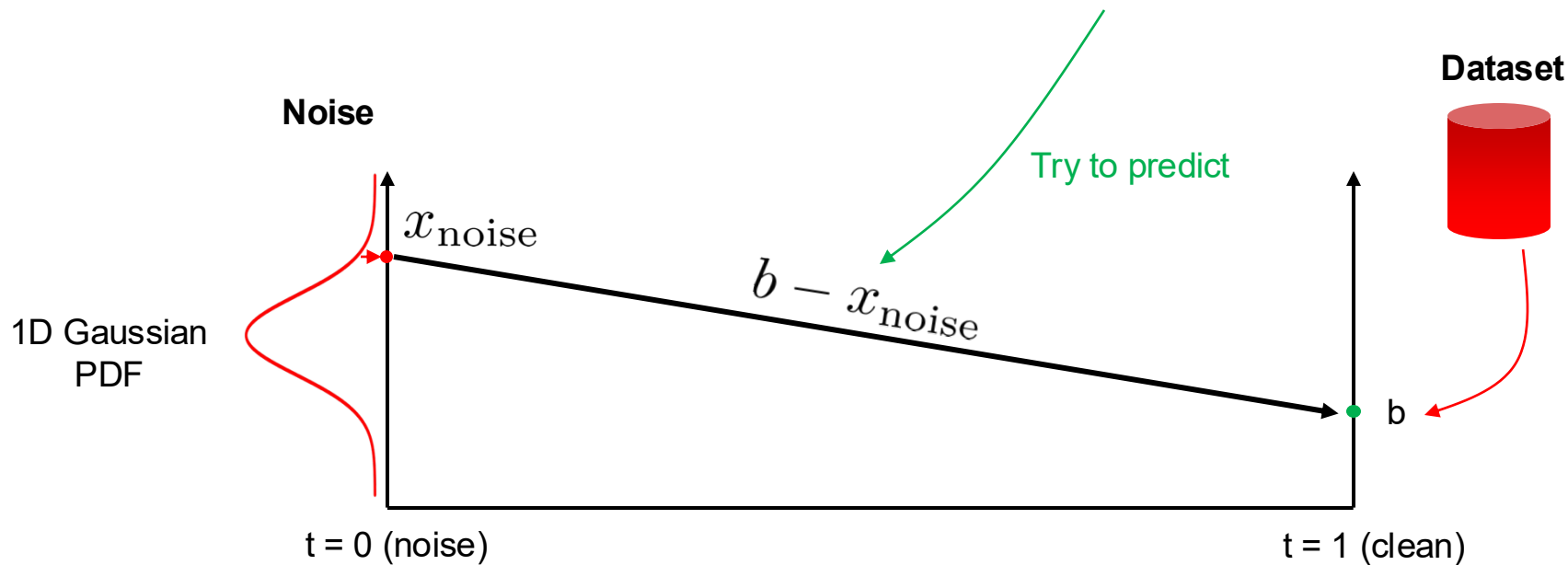
Sample noise, data, optimize

$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$



Sample noise, data, optimize

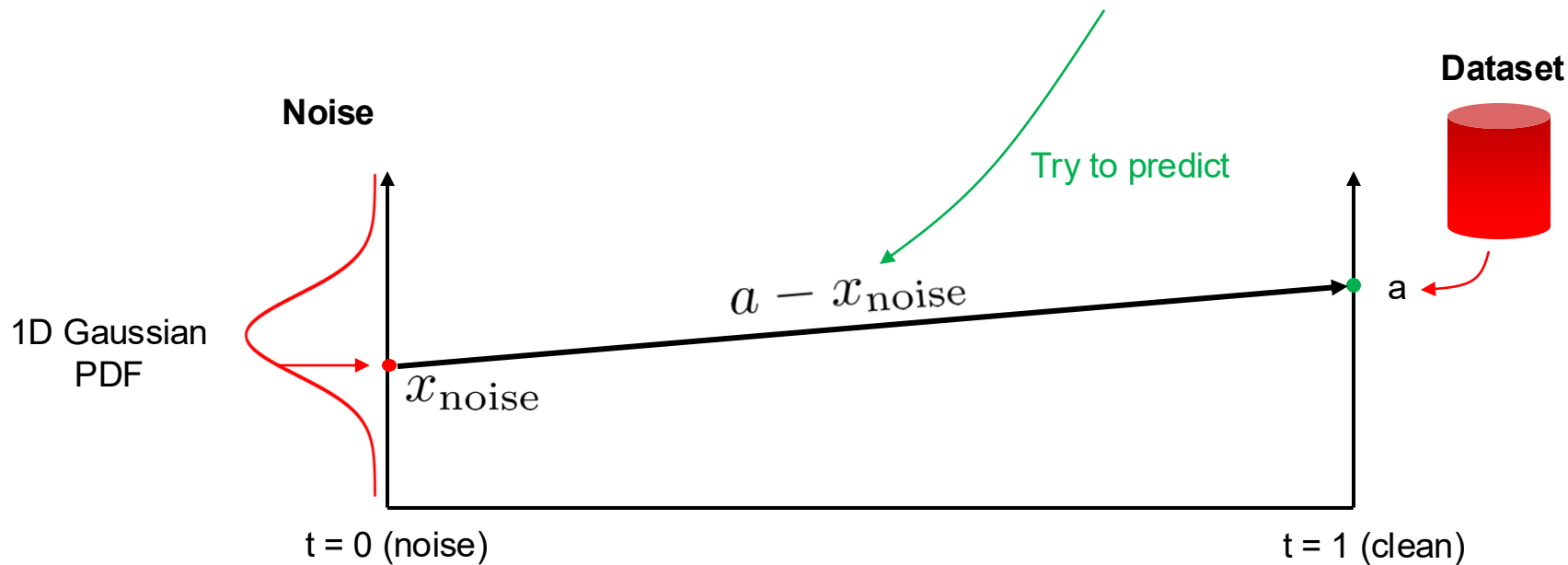
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$



*Sample many times...

Sample noise, data, optimize

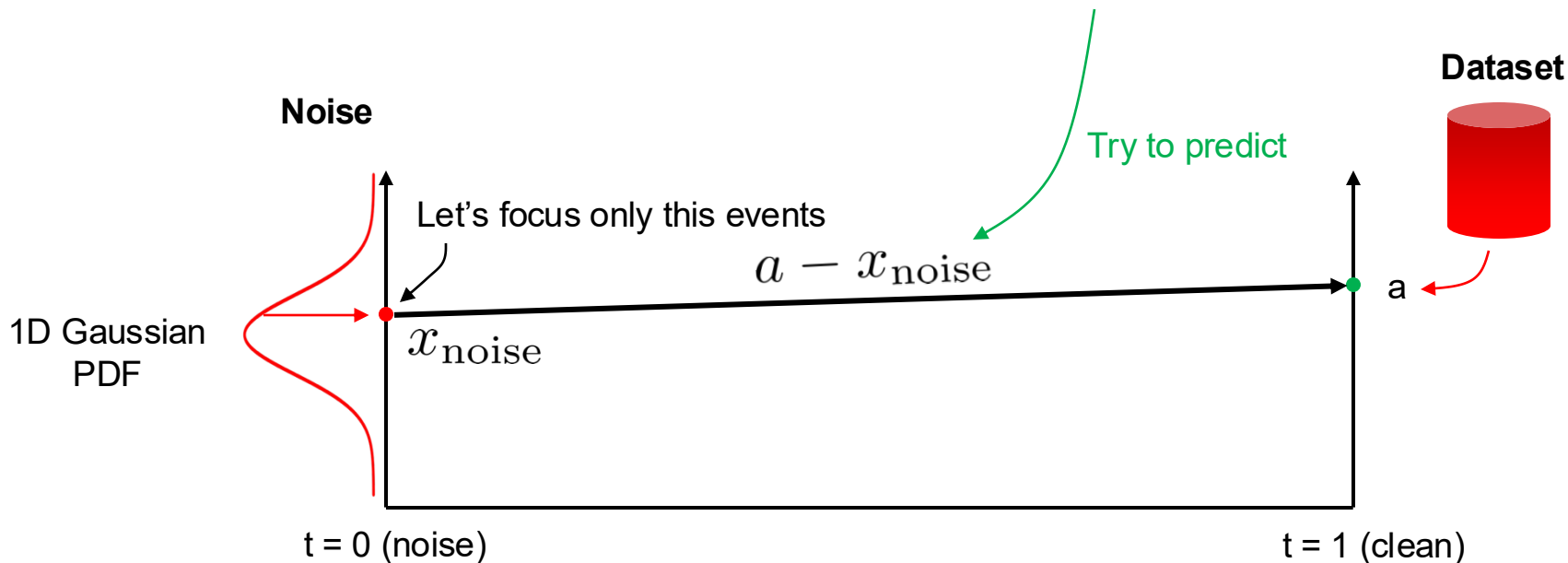
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$



*Sample many times...

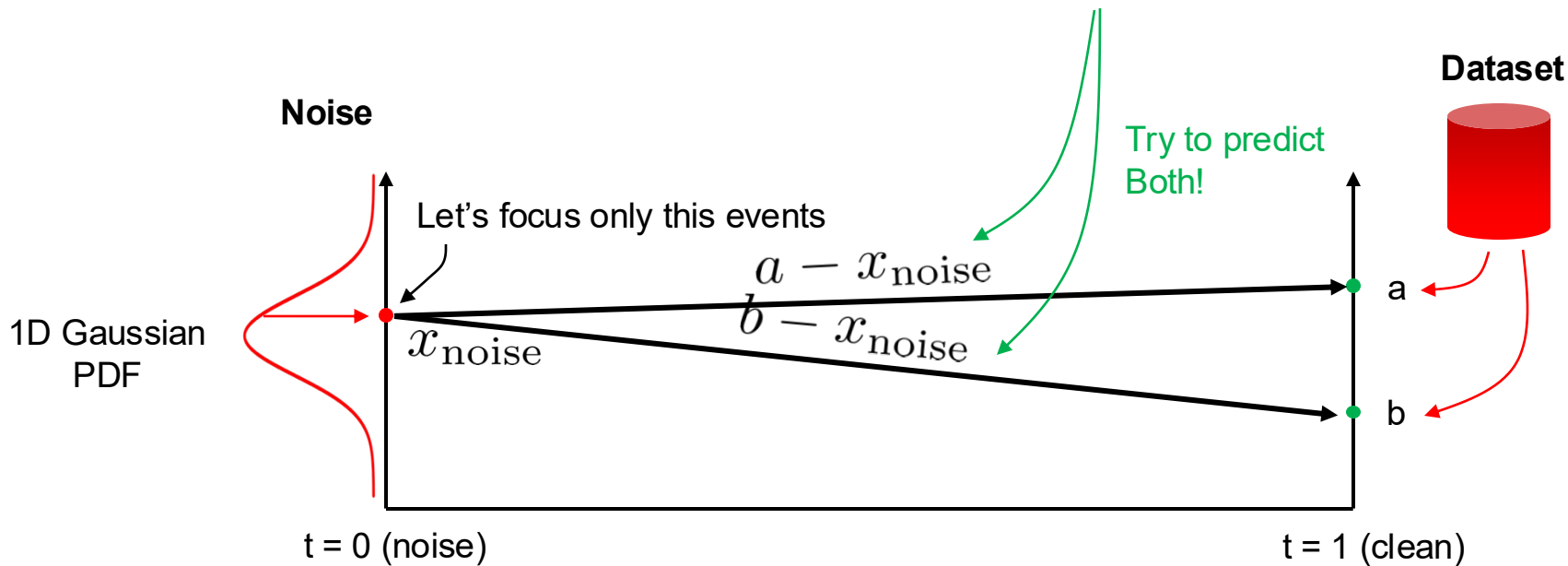
Let's focus on one event!

$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$



There are two possible outcomes!

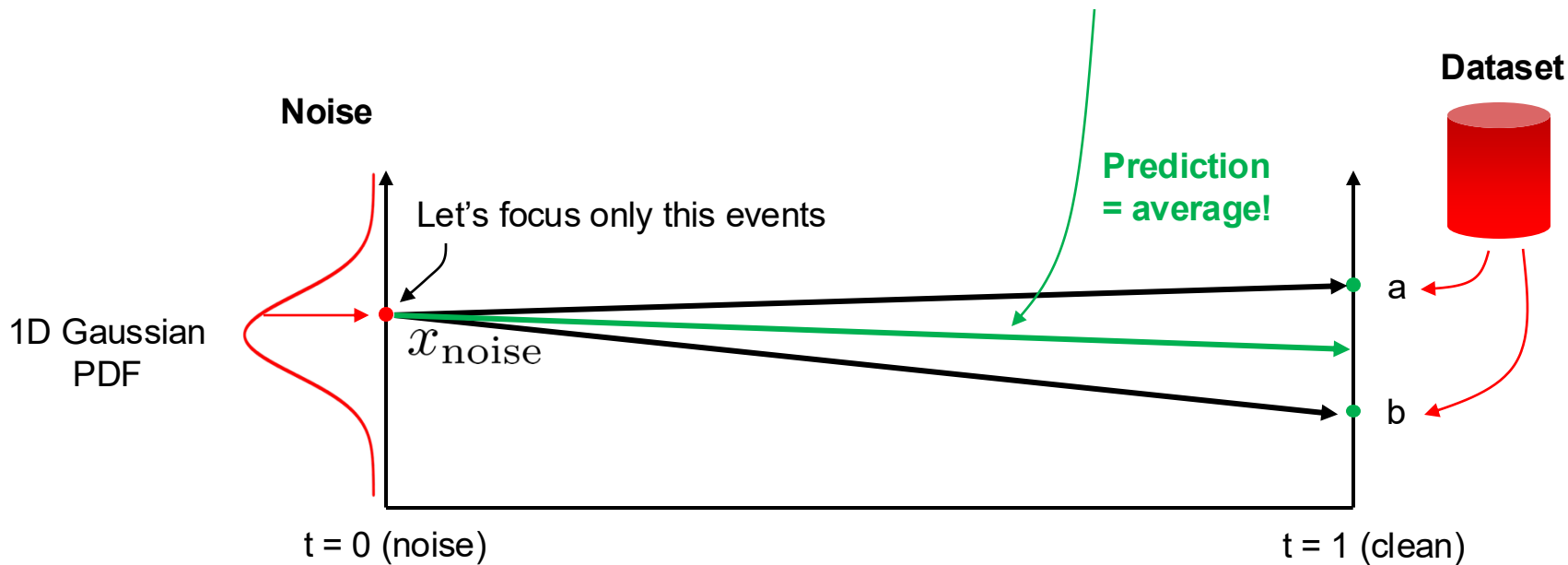
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$



*Can neural net predict two values?

What will neural net predict? (Marginal flow)

$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_{\text{noise}})||^2$$

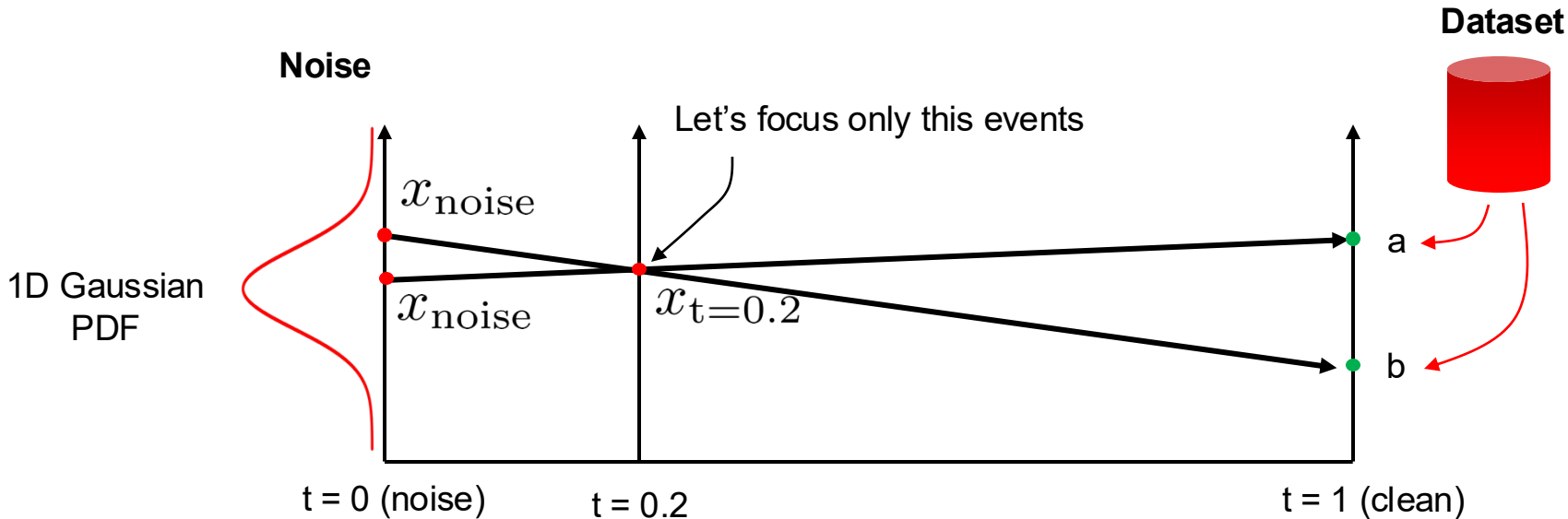


*At noise, the prediction is always the average!

Let's try at $t > 0$

What will neural net predict? (Marginal flow)

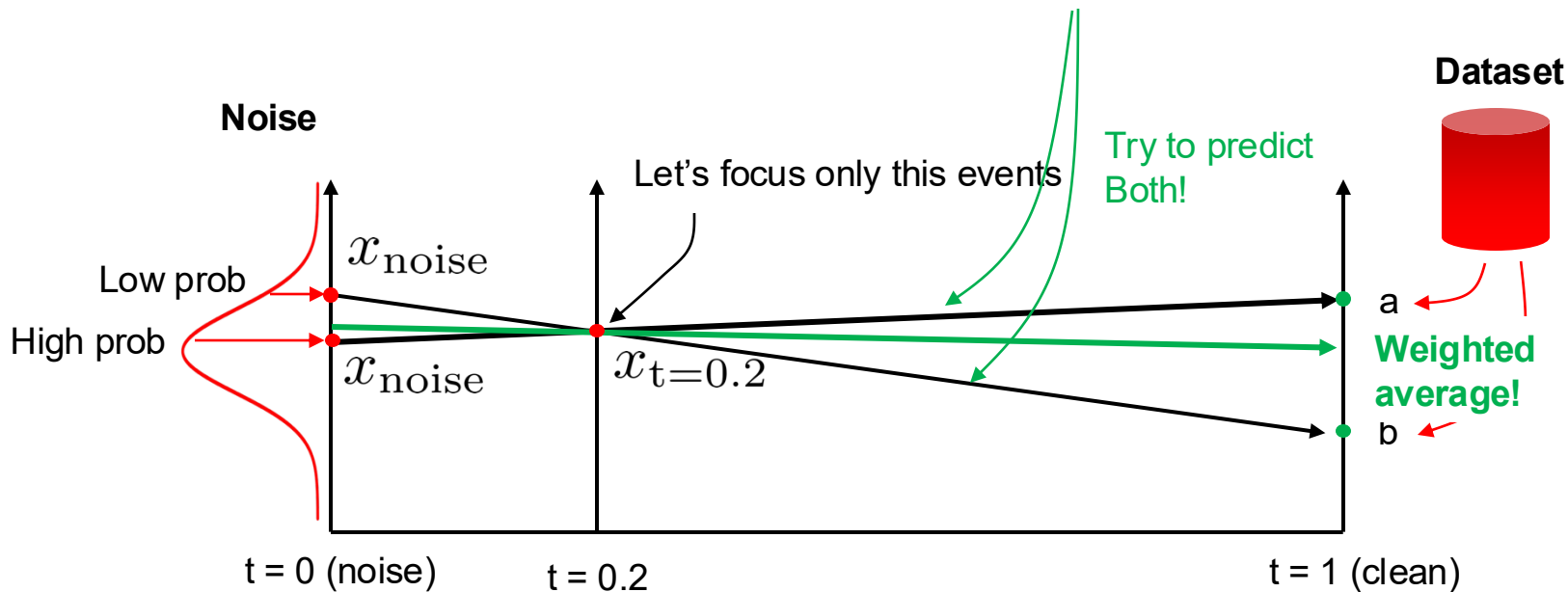
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_t)||^2$$



Don't forget: $x_t = tx_{\text{clean}} + (1 - t)x_{\text{noise}}$

What will neural net predict? (Marginal flow)

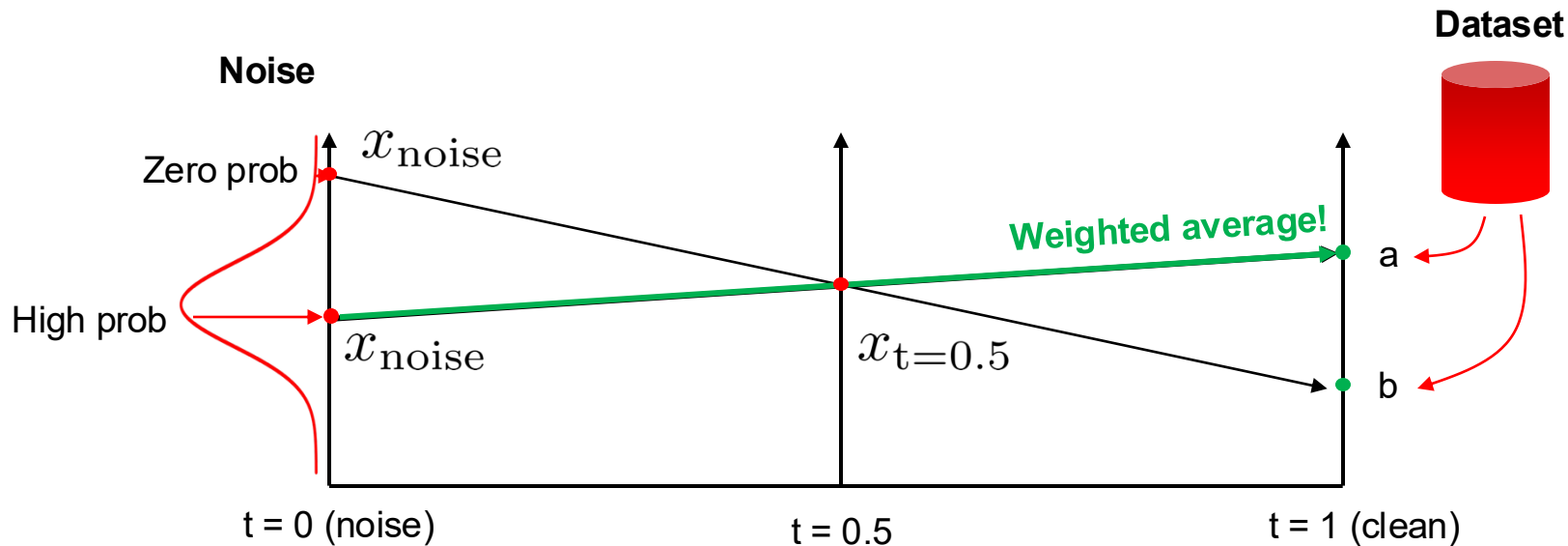
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_t)||^2$$



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What will neural net predict? (Marginal flow)

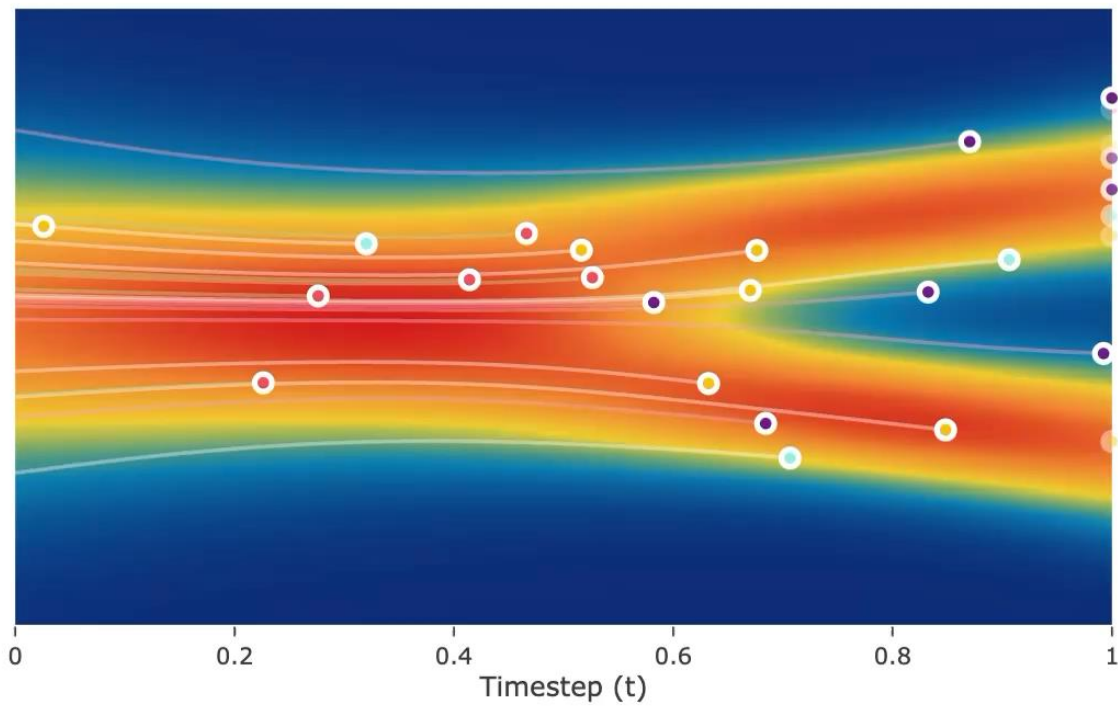
$$L = ||(x_{\text{clean}} - x_{\text{noise}}) - u_{\theta}(x_t)||^2$$



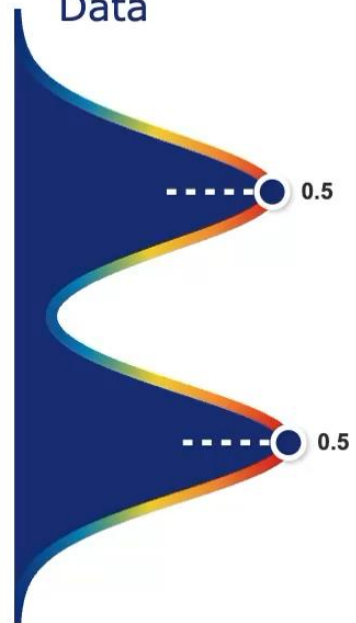
Don't forget: $x_t = tx_{\text{clean}} + (1 - t)x_{\text{noise}}$

*As t moves closer to clean, it is biased more to a real data point.

Prior

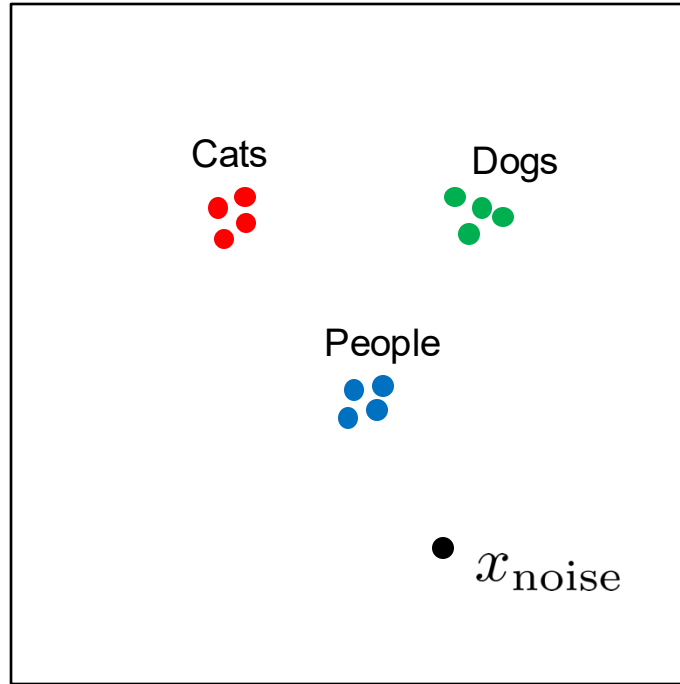


Data

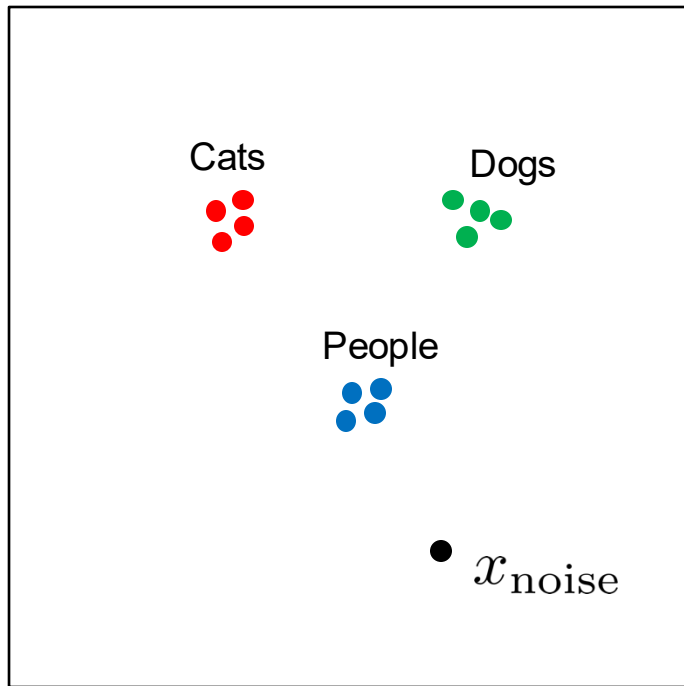


Classifier-free guidance (CFG)

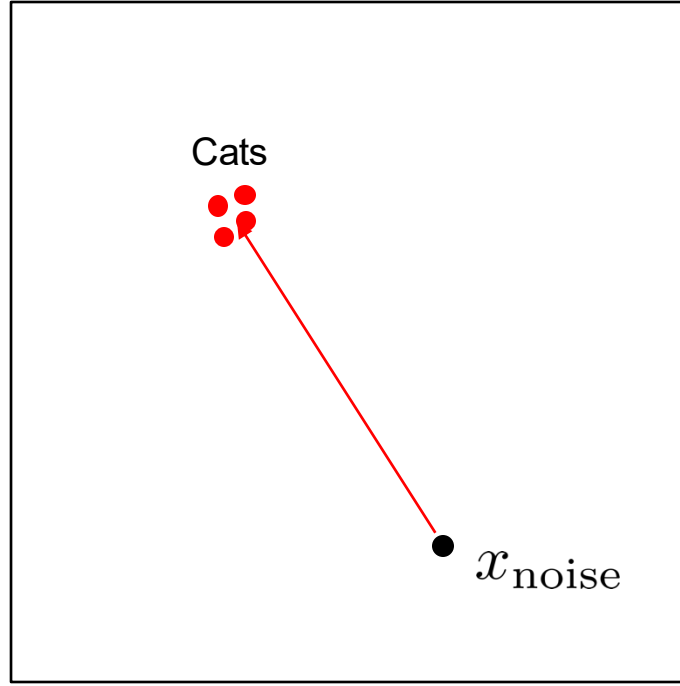
Now, imagine 2D world with 3 types of points



I want a "Cats" model, what's the marginal flow?

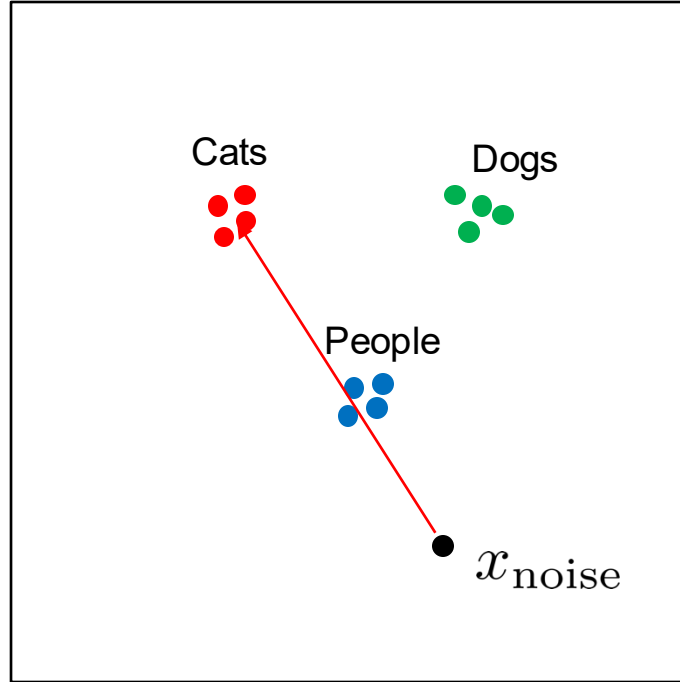


I want a "Cats" model, what's the marginal flow?



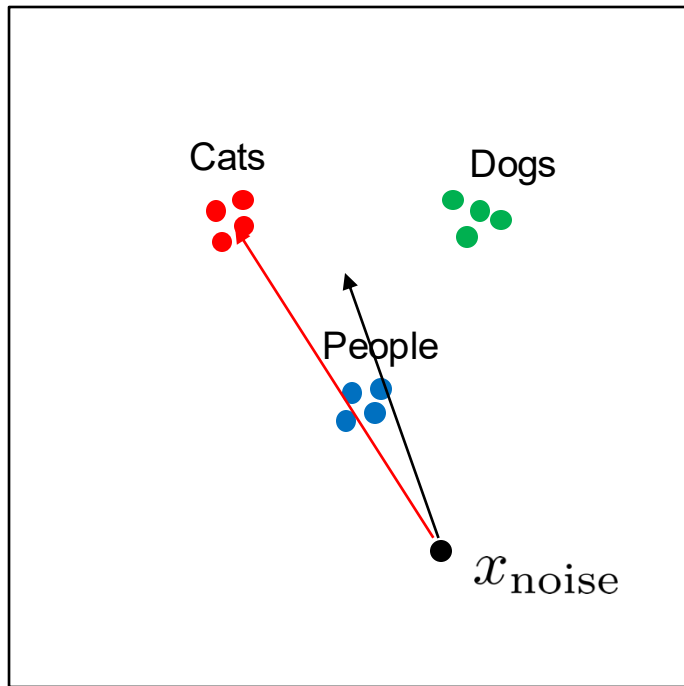
*At noise, point to the average!

What's unconditional marginal flow?



*At noise, point to the average!

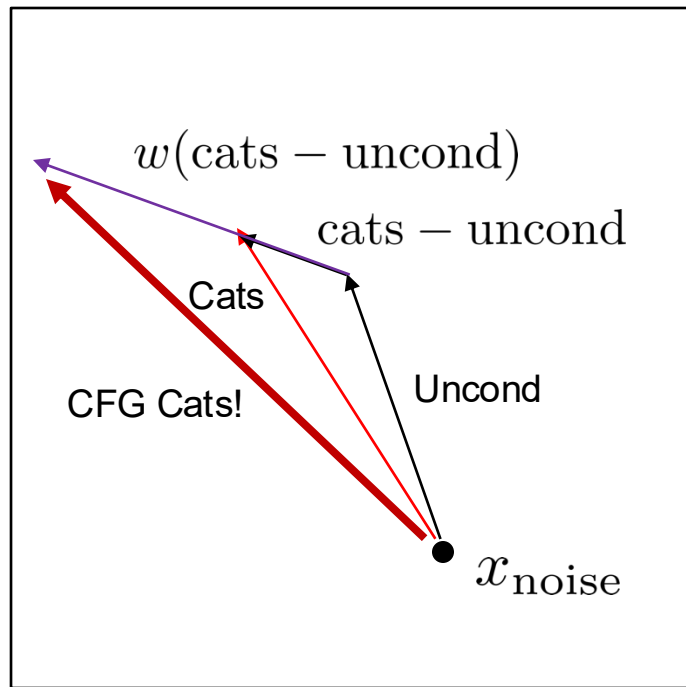
What's unconditional marginal flow?



*At noise, point to the average!

CFG = Exaggerate the "class" more

$$\text{CFG} = \text{uncond} + w(\text{cats} - \text{uncond})$$



*Follow the new direction

Let's do worksheets!